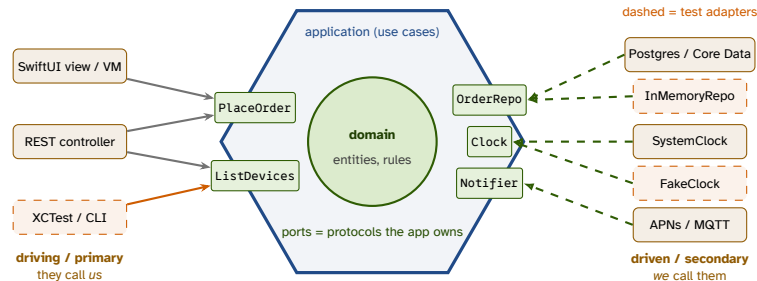
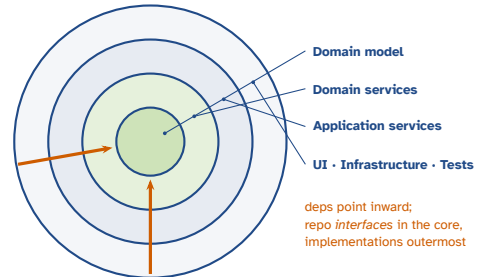


In one line: Put the **domain in the middle** and make everything else — UI, HTTP, database, clock, push — a replaceable **adapter** plugged into a **port** the application defines. Hexagonal (Cockburn 2005), Onion (Palermo 2008) and Clean (Martin 2012) are *one* dependency rule in three vocabularies: **source dependencies point inward**. A **modular monolith** applies the same boundaries between modules in one deployable, before (or instead of) paying for microservices.

Ports & adapters



Onion rings



How it works

- **Port** = an interface at the application boundary, named in domain terms (OrderRepository, not PostgresDAO). **Adapter** = code that converts between a technology and a port.
- **Driving (primary) ports** — what the app *offers*: use-case interfaces called by UI, HTTP, CLI, tests. **Driven (secondary) ports** — what the app *needs*: persistence, messaging, time, payments. The app declares them; infrastructure implements them (DIP).
- Cockburn's goal: the app can be **driven equally by users, programs, automated tests or batch scripts**, and developed/tested **in isolation** from its devices and databases. Six sides mean nothing — just room to draw ports.
- **Onion** — concentric rings; the core has no outward deps; Palermo stresses that **infrastructure is outer**, not the bottom layer (a classic 3-tier stack has UI → BLL → DAL, so the domain depends on the DB).
- **Relation:** same rule, different emphasis — hexagonal: inside vs outside + symmetric left/right; onion: rings inside the core; Clean: named rings + the **Dependency Rule** (see coordinator-repository-di-clean).
- **Testability payoff:** the domain + use cases run in unit tests with an InMemoryRepo and a FakeClock — no simulator, DB or network; same use case, adapter swapped at the composition root.

Example — a use case with two driven ports

```
protocol DeviceRepository { func all() async throws -> [Device] }
protocol Clock { func now() -> Date } // driven ports
struct ListStateDevices { // driving port
    let repo: DeviceRepository; let clock: Clock
    func callAsFunction() async throws -> [Device] {
        let now = clock.now()
        return try await repo.all().filter {
            now.timeIntervalSince($0.lastSeen) > 900 } } }
// adapters live in OTHER modules:
struct APIDeviceRepository: DeviceRepository { /* URLSession */ }
struct FakeClock: Clock { var t: Date; func now() -> Date { t } }
```

On iOS

Feature SPM packages expose a small public API (driving port) and declare what they need as protocols (driven ports: DeviceRepository, Analytics, Clock); a Networking package adapts URLSession to them; the **app target** is the composition root wiring adapters in. The app *is* a modular monolith: one binary, compiler-enforced boundaries (internal vs public). See modularization-spm.

Modular monolith vs microservices

	Modular monolith	Microservices
deploy boundary	one unit, one version module API, enforced by build/lint	independent per service network API — hard, can't cheat
calls	in-process, typed, free	network: latency, partial failure
data	one DB, schema per module	DB per service
consistency	local ACID transactions	sagas + outbox; eventual
ops	one pipeline, one log	tracing, discovery, versioning
scales	whole app	per service, per team

When to split a module out: it needs **independent deploy cadence** (a team blocked by others' releases), **different scaling** or runtime, **fault isolation**, or a hard security boundary. Not because "it's cleaner" — a bad boundary across a network is a **distributed monolith**. Fowler's *MonolithFirst*: find the boundaries in-process, then extract. Modules talk via public APIs or in-process events, never another module's tables.

Traps

- ! "Hexagonal = six layers" — no; two sides (driving/driven), N ports.
- ! The *port* belongs to the app, not to the adapter; an interface generated from the DB or SDK is not a port.
- ! A port per class "for testability" is ceremony; ports go at I/O boundaries. A CRUD screen needs no hexagon.
- ! Microservices don't fix a tangled model — they distribute it.

Remember

App owns the ports; adapters plug in on both sides; arrows point in; split the deploy only when a boundary has earned it.